

DSA Day 9: Circular Linked List

Book: Data Structures Through C in Depth - Ch3 §3.3-3.4 · pp 72-83 | Code in Python

Overview

The last node points back to the first. Great for round-robin. A header/sentinel node simplifies edge cases.

Key concepts to master

- last.next = head
- Stop traversal when you return to head
- Header/sentinel node

Python implementation

```
def traverse_circular(head):  
    if not head: return  
    cur=head  
    while True:  
        print(cur.data)  
        cur=cur.next  
        if cur is head: break
```

Complexity

Same as singly linked; traversal $O(n)$.

Common pitfalls

- Infinite loop if stop condition is wrong
- Updating last.next on insert/delete at head

Practice

- Build a 4-node circle and traverse once
- Delete one node safely

